

# PAPER\_242 — Rings of Relativity: Einstein Ring Lensing Amplification in the Full MUGE

## GAL-CLUS-022058s — Static Einstein-Ring Lensing Factor in the Master Universal Gravity Equation

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**Framework:** Unified Quantum Field Framework (UQFF) v4.10

**Calculator Class:** RingsOfRelativityEinsteinLensingMUGECalculator (CP3, Session 60)

**Encoded By:** Grok (xAI), October 2025 (C++ source); Python CP3 integration March 2026

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### Abstract

This paper presents the full Master Universal Gravity Equation (MUGE) for the GAL-CLUS-022058s gravitational lens system (“Rings of Relativity”). The key unique mathematical contribution is the **static Einstein ring lensing amplification factor**  $L_t$ , derived from the angular diameter distance ratio  $D_{LS}/D_S$  and the Schwarzschild radius of the galaxy cluster lens, applied as a multiplicative correction to the base gravitational field.

This is expressly **distinct** from the dynamic lensing modulation  $L(t) = L_0 e^{-t/\tau} \cos(\omega t)$  already captured in CP3 class 81 (UQFFLensingModulationRingsCalculator, Session 53). The present paper captures the static, geometry-driven Einstein ring correction.

## 2. System Parameters

| Parameter              | Symbol                           | Value                  | Units            |
|------------------------|----------------------------------|------------------------|------------------|
| Lens mass              | $M$                              | $10^{14} M_{\odot}$    | kg               |
| Einstein radius        | $r$                              | $3.086 \times 10^{20}$ | m (~10 kpc)      |
| Redshift of lens       | $z_{\text{lens}}$                | 0.5                    | dimensionless    |
| Angular distance ratio | $L_{\text{factor}} = D_{LS}/D_S$ | 0.67                   | dimensionless    |
| Magnetic field         | $B$                              | $10^{-5}$              | T                |
| Gas velocity           | $v_{\text{gas}}$                 | $10^5$                 | m/s              |
| Oscillatory amplitude  | $A_{\text{osc}}$                 | $10^{-12}$             | m/s <sup>2</sup> |

## 3. The Novel Lensing Amplification Term

### 3.1 Einstein Ring Lensing Factor

The static gravitational lensing amplification of the measured gravitational field along the line of sight through the Einstein ring geometry:

$$L_t = \frac{GM}{c^2 r} \cdot L_{\text{factor}}, \quad L_{\text{factor}} = \frac{D_{LS}}{D_S} = 0.67$$

$$\text{corr}_L = 1 + L_t$$

where: -  $GM/c^2r$  is the dimensionless Schwarzschild-radius-to-Einstein-radius ratio -  $D_{LS}$  = angular diameter distance from lens to source -  $D_S$  = angular diameter distance from observer to source - The factor  $D_{LS}/D_S$  encodes the lensing geometry in the single-lens single-source approximation

### 3.2 Friedmann Correction at $z = 0.5$

$$H(z = 0.5) = H_0 \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda} = H_0 \sqrt{0.3 \cdot (1.5)^3 + 0.7}$$

This mid-redshift Friedmann correction is distinct from  $z = 0$  (local),  $z = 3.5$  (HUDF), or  $z = 0.01$  (nearby lenses) used in earlier classes.

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## 4. Full MUGE: All Nine Terms

$$g_{\text{Rings}}(r, t) = \underbrace{T_1}_{\text{base+H+B+L}} + \underbrace{T_2}_{\text{UQFF}} + \underbrace{T_3}_{\Lambda} + \underbrace{T_4}_{\text{EM}} + \underbrace{T_q}_{\text{QM}} + \underbrace{T_{\text{fl}}}_{\text{fluid}} + \underbrace{T_{\text{osc}}}_{\text{2-mode}} + \underbrace{T_{\text{DM}}}_{\text{DM+pert2}} + \underbrace{T_{\text{wind}}}_{\text{stellar wind}}$$

**Term 1 — Base gravity with  $H(z)$ , magnetic, Einstein lensing:**

$$T_1 = \frac{GM}{r^2} (1 + H(z)t) (1 - B/B_{\text{crit}}) (1 + L_t)$$

**Term 2 — UQFF unified field with time-reversal:**

$$T_2 = (U_{g1} + U_{g4})(1 + f_{TRZ})$$

**Term 3 — Cosmological constant:**

$$T_3 = \frac{\Lambda c^2}{3}$$

**Term 4 — Electromagnetic with vacuum density ratio:**

$$T_4 = \frac{qv_{\text{gas}}B}{m_p} \left( 1 + \frac{\rho_{\text{UA}}}{\rho_{\text{SCm}}} \right) \cdot s_{\text{EM}}, \quad \frac{\rho_{\text{UA}}}{\rho_{\text{SCm}}} = \frac{7.09 \times 10^{-36}}{7.09 \times 10^{-37}} = 10$$

**Term 5 — Quantum uncertainty:**

$$T_q = \frac{\hbar}{\sqrt{\Delta x \Delta p}} \psi \frac{2\pi}{t_H}$$

**Term 6 — Fluid (buoyancy-acceleration):**

$$T_{\text{fl}} = \frac{\rho_{\text{fluid}} V g_{\text{base}}}{M}$$

**Term 7 — Two-mode oscillation (standing wave + traveling wave):**

$$T_{\text{osc}} = 2A \cos(kx) \cos(\omega t) + \frac{2\pi}{t_{\text{Gyr}}} A \cos(kx - \omega t)$$

The first mode is a **standing wave decomposition** ( $2 \cos \cos$ ); the second mode is a **Gyr-scaled traveling wave**. Together they form a beating-mode pair unique to this system.

**Term 8 – Dark matter with  $\delta_2 = 3GM/r^3$  perturbation:**

$$T_{\text{DM}} = \frac{(M + M_{\text{DM}})(\delta\rho/\rho + 3GM/r^3)}{M}$$

The second-order perturbation  $3GM/r^3$  is a tidal-force density correction distinct from density-contrast  $\delta\rho/\rho$ .

**Term 9 – Stellar wind feedback:**

$$T_{\text{wind}} = \frac{\rho_{\text{wind}} v_{\text{wind}}^2}{\rho_{\text{fluid}}}$$

## 5. Distinction from Prior Art

| Feature         | Class 81 (Session 53)                                  | This class (Session 60)                     |
|-----------------|--------------------------------------------------------|---------------------------------------------|
| Lensing factor  | $L(t) = L_0 e^{-t/\tau} \cos(\omega_{\text{lens}} t)$  | $L_t = (GM/c^2 r) \cdot (D_{LS}/D_S)$       |
| Physics         | Dynamic lens transit alignment                         | Static Einstein ring geometry               |
| Time dependence | Yes (decaying oscillation)                             | No (geometric constant)                     |
| Parameter       | $L_0 = 0.15, \tau_{\text{lens}}, \omega_{\text{lens}}$ | $L_{\text{factor}} = 0.67$ (distance ratio) |
| Oscillation     | None                                                   | Two-mode standing + traveling wave          |
| DM pert2        | Not present                                            | $3GM/r^3$ tidal correction                  |
| Wind term       | Not present                                            | $\rho_w v_w^2 / \rho_{\text{fl}}$           |

## 6. Numerical Verification

Using default parameters ( $M = 1.989 \times 10^{44}$  kg,  $r = 3.086 \times 10^{20}$  m,  $z = 0.5$ ):

$$L_t = \frac{(6.6743 \times 10^{-11})(1.989 \times 10^{44})}{(2.998 \times 10^8)^2 (3.086 \times 10^{20})} \times 0.67 \approx 1.6 \times 10^{-3}$$

$$\text{corr}_L \approx 1.0016 \quad (\sim 0.16\% \text{ amplification})$$

$$H(z = 0.5) = H_0 \sqrt{0.3 \times 3.375 + 0.7} \approx 1.27 H_0$$

The lensing correction is small but non-zero and physically motivated by the Einstein ring geometry of GAL-CLUS-022058s ( $\theta_E \approx 10''$ , confirmed by HST imaging).

## 7. Integration Into UQFF Pipeline

- **CP3 class:** RingsOfRelativityEinsteinLensingMUGECalculator (112th calculator, Session 60)
  - **Aggregator:** v2.4.0 — registered in CP3\_CALCULATORS
  - **Source:** Doc 8 C++ class RingsOfRelativity (Grok/xAI, October 2025)
  - **Complementary classes:**
    - Class 81 UQFFLensingModulationRingsCalculator — dynamic lens transit
    - Class 111 NGC3603FullMUGECavityPressureCalculator — companion new class (PAPER\_243)
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## 8. References

- Hubble Space Telescope imaging of GAL-CLUS-022058s (Infante et al. 2022)
  - Einstein (1936): Lens-star problem, Science 84, 506
  - Schneider, Ehlers & Falco (1992): Gravitational Lenses, Springer
  - Murphy, D.T. (2025): UQFF Manuscript — Doc 8 (Rings of Relativity MUGE, October 2025)
  - Grok (xAI): C++ class RingsOfRelativity, encoded October 2025
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*PAPER\_242 | Session 60 | CP3 class 111 (RingsOfRelativityEinsteinLensingMUGECalculator) | UQFF v4.10*